

BOOK 5 — AI ENGINEERING

Become an AI-Native Engineer

Dedicated Detailed Curriculum

PURPOSE OF THIS BOOK

This book is NOT designed to make you:

- an ML researcher,
- a prompt engineer only,
- or someone who simply calls AI APIs.

This book IS designed to help you become:

A Strong AI-Native Systems Engineer

Meaning:

- you understand how modern AI systems work,
 - you can integrate AI into production systems,
 - you can design reliable AI workflows,
 - you can debug AI systems,
 - you can operate AI-powered platforms,
 - and you can think architecturally about AI.
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WHO THIS BOOK IS FOR

This roadmap is optimized for engineers who already have:

- software engineering background,
- Linux familiarity,
- backend understanding,
- debugging mindset,
- infrastructure exposure.

Which means: You are already in a strong position.

CORE PHILOSOPHY

Do NOT become:

- overly dependent on AI,
- shallow in understanding,
- obsessed with hype.

Become:

- capable of building intelligent systems,
 - capable of integrating AI reliably,
 - capable of operating AI in production.
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THE AI ENGINEERING STACK

This book is divided into 8 PARTS.

Part	Focus
Part 1	AI Foundations
Part 2	LLM Engineering
Part 3	Prompt & Context Engineering
Part 4	RAG Systems
Part 5	AI Agents & Workflows
Part 6	AI Infrastructure
Part 7	AI Reliability & Evaluation
Part 8	AI Systems Architecture

YEARLY EXECUTION STRATEGY

YEAR 1

Focus:

- foundations,
- LLM engineering,
- RAG,
- AI integration,
- practical systems.

YEAR 2

Focus:

- agents,
- AI infrastructure,
- reliability,
- orchestration,
- architecture,
- production AI systems.

WEEKLY EXECUTION MODEL

Thursday — Primary AI Day

Core Study

60–90 mins

Build/Experiment

60 mins

Notes/Debugging

20–30 mins

Optional Daily AI Habit

15–20 mins/day:

- AI engineering blog,
 - GitHub repo reading,
 - architecture discussion,
 - paper summary,
 - prompt experimentation.
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PART 1 — AI FOUNDATIONS

Goal

Understand AI concepts deeply enough to build systems confidently.

CHAPTER 1 — AI Landscape

Topics

- AI vs ML vs Deep Learning
- Generative AI overview
- LLM ecosystem
- Open-source vs closed-source models
- Modern AI industry direction

Outcome

You understand where AI engineering fits.

CHAPTER 2 — Neural Network Intuition

Topics

- What neural networks do
- Training intuition
- Weights/biases intuition
- Why deep learning works
- High-level transformer intuition

Important

No heavy math required initially.

CHAPTER 3 — LLM Fundamentals

Topics

- Tokens
- Tokenization
- Context windows
- Next-token prediction

- Temperature/top_p
- Hallucinations
- Reasoning limitations
- Context management

Outcome

You understand LLM behavior practically.

CHAPTER 4 — Embeddings

Topics

- Semantic similarity
- Vector representations
- Embedding generation
- Search intuition
- Retrieval concepts

Build

Mini semantic search engine.

CHAPTER 5 — Transformers Intuition

Topics

- Attention intuition
- Transformer architecture overview
- Why transformers changed AI
- Scaling behavior

Goal

Conceptual understanding.

PART 1 PROJECTS

Mini Projects

- Token counter tool
- Semantic similarity search
- Embedding explorer

PART 2 — LLM ENGINEERING

Goal

Learn how to build applications using modern LLMs.

CHAPTER 1 — OpenAI APIs

Topics

- Chat completions
- Structured outputs
- JSON mode
- Function/tool calling
- Streaming responses
- Cost optimization
- Retry handling

Build

AI-powered API service.

CHAPTER 2 — Open-Source Models

Topics

- Ollama
- Hugging Face
- Local models
- Model selection
- Model tradeoffs

Build

Run local LLM on your machine.

CHAPTER 3 — Prompt Engineering

Topics

- Prompt structure
- System prompts
- Few-shot prompting
- Role prompting
- Prompt chaining
- Guardrails
- Output formatting

Important

Understand deeply. Do NOT become prompt-only engineer.

CHAPTER 4 — Structured AI Systems

Topics

- Pydantic + LLM outputs
- Schema validation
- Deterministic workflows
- AI orchestration patterns

Build

Reliable AI extraction pipeline.

CHAPTER 5 — Tool Calling

Topics

- Function calling
- External tools
- API integrations
- Tool orchestration
- Agent/tool boundaries

Build

AI assistant with real tools.

PART 2 PROJECTS

Projects

- AI CLI assistant
 - AI shell assistant
 - AI code summarizer
 - AI API backend
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PART 3 — PROMPT & CONTEXT ENGINEERING

Goal

Learn how to control AI system behavior effectively.

CHAPTER 1 — Context Engineering

Topics

- Context window management
 - Context injection
 - Memory handling
 - Compression strategies
 - Retrieval optimization
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CHAPTER 2 — Prompt Architectures

Topics

- Multi-step prompting
 - Reflection prompting
 - Planning prompts
 - Chain-of-thought intuition
 - Routing prompts
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CHAPTER 3 — AI Workflow Design

Topics

- AI pipelines
 - Validation stages
 - Human-in-loop systems
 - Reliability layering
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CHAPTER 4 — Safety & Control

Topics

- Prompt injection
 - Data leakage prevention
 - Security basics
 - Output constraints
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PART 3 PROJECTS

Projects

- Multi-stage AI workflow
 - AI document processor
 - Controlled AI assistant
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PART 4 — RAG SYSTEMS

Goal

Master Retrieval-Augmented Generation deeply.

CHAPTER 1 — RAG Foundations

Topics

- Why RAG exists
- Retrieval flow
- Chunking

- Search pipelines
 - Grounding AI responses
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CHAPTER 2 — Vector Databases

Topics

- ChromaDB
 - FAISS basics
 - Pinecone overview
 - Similarity search
 - Indexing strategies
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CHAPTER 3 — Retrieval Optimization

Topics

- Chunk size tradeoffs
 - Metadata filtering
 - Hybrid search
 - Re-ranking
 - Retrieval quality
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CHAPTER 4 — Production RAG

Topics

- RAG observability
 - Latency optimization
 - Hallucination reduction
 - Document pipelines
 - Incremental indexing
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CHAPTER 5 — Advanced RAG

Topics

- Agentic RAG
- Graph RAG intuition
- Multi-source retrieval
- Long-context systems

PART 4 PROJECTS

Projects

- Engineering documentation chatbot
 - Internal company knowledge system
 - GitHub repository assistant
 - AI troubleshooting assistant
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PART 5 — AI AGENTS & WORKFLOWS

Goal

Learn orchestration and autonomous AI systems.

CHAPTER 1 — Agent Fundamentals

Topics

- What agents are
 - Agent loops
 - Planning
 - Tool usage
 - Decision flow
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CHAPTER 2 — Agent Architectures

Topics

- Single-agent systems
 - Multi-agent systems
 - Supervisory agents
 - Workflow agents
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CHAPTER 3 — Workflow Automation

Topics

- AI pipelines
 - Event-driven AI systems
 - Automation orchestration
 - AI task routing
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CHAPTER 4 — Memory Systems

Topics

- Conversation memory
 - Long-term memory
 - Retrieval memory
 - State handling
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CHAPTER 5 — Agent Reliability

Topics

- Failure loops
 - Tool misuse
 - Infinite loops
 - Agent observability
 - Human override systems
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PART 5 PROJECTS

Projects

- AI DevOps assistant
 - AI deployment analyzer
 - AI workflow orchestrator
 - Autonomous debugging assistant
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PART 6 — AI INFRASTRUCTURE

Goal

Operate AI systems like a production engineer.

CHAPTER 1 — Model Serving

Topics

- API serving
 - vLLM overview
 - Ollama serving
 - TGI overview
 - Inference APIs
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CHAPTER 2 — GPU Basics

Topics

- GPU intuition
 - VRAM basics
 - Quantization intuition
 - Model size tradeoffs
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CHAPTER 3 — AI Deployment

Topics

- Dockerized AI apps
 - Kubernetes deployment
 - Scaling inference services
 - Load balancing AI APIs
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CHAPTER 4 — AI Observability

Topics

- AI metrics

- Token monitoring
 - Latency tracking
 - Cost monitoring
 - Failure analysis
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CHAPTER 5 — AI Platform Engineering

Topics

- Multi-model systems
 - AI gateways
 - Routing systems
 - AI platform architecture
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PART 6 PROJECTS

Projects

- Self-hosted AI platform
 - AI gateway service
 - AI inference deployment system
 - Kubernetes AI deployment stack
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PART 7 — AI RELIABILITY & EVALUATION

Goal

Build trustworthy AI systems.

CHAPTER 1 — Evaluation Fundamentals

Topics

- AI evaluation basics
 - Accuracy vs usefulness
 - Hallucination measurement
 - Benchmark intuition
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CHAPTER 2 — AI Reliability

Topics

- Failure handling
 - Retry systems
 - Confidence scoring
 - Validation systems
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CHAPTER 3 — AI Guardrails

Topics

- Input validation
 - Output validation
 - Toxicity handling
 - Safety filters
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CHAPTER 4 — Human-in-Loop Systems

Topics

- Human review pipelines
 - Escalation systems
 - Hybrid workflows
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CHAPTER 5 — Cost & Performance Optimization

Topics

- Token optimization
 - Caching
 - Context compression
 - Latency optimization
 - Model routing
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PART 7 PROJECTS

Projects

- AI evaluation dashboard
 - AI guardrail system
 - AI quality analyzer
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PART 8 — AI SYSTEMS ARCHITECTURE

Goal

Think like an AI systems architect.

CHAPTER 1 — AI System Design

Topics

- AI architecture patterns
 - AI microservices
 - AI orchestration layers
 - Event-driven AI systems
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CHAPTER 2 — Enterprise AI Systems

Topics

- Internal copilots
 - Enterprise RAG
 - AI knowledge systems
 - AI workflow platforms
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CHAPTER 3 — AI + Distributed Systems

Topics

- Distributed inference
- Queue-based AI systems
- Streaming AI systems

- AI scaling patterns
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CHAPTER 4 — AI Reliability at Scale

Topics

- Large-scale monitoring
 - AI incident handling
 - Operational reliability
 - Multi-region AI systems
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CHAPTER 5 — Future AI Engineering Trends

Topics

- Agentic systems
 - AI-native infrastructure
 - AI operating systems
 - Long-context systems
 - AI platform evolution
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PART 8 CAPSTONE PROJECT

Build Your Signature System

Recommended:

AI Operations Platform

Capabilities:

- log analysis,
- deployment intelligence,
- AI copilots,
- observability,
- incident diagnosis,
- automated remediation,
- distributed architecture,
- Kubernetes deployment,
- RAG,
- AI agents.

This becomes:

- your architect-level showcase,
 - your portfolio centerpiece,
 - your long-term engineering identity.
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LEARNING PRIORITY ORDER

PHASE 1

Foundations

- Part 1
- Part 2

Duration: 2–3 months

PHASE 2

Practical AI Systems

- Part 3
- Part 4

Duration: 3–4 months

PHASE 3

AI Automation & Infrastructure

- Part 5
- Part 6

Duration: 4–5 months

PHASE 4

Reliability & Architecture

- Part 7
- Part 8

Duration: 4–6 months

WEEKLY EXECUTION EXAMPLE

Thursday

AI Study Day

30 mins

Concepts

60 mins

Hands-on implementation

30 mins

Debugging/notes

Saturday

AI Project Integration

Build:

- real features,
 - integrations,
 - infrastructure,
 - orchestration.
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Sunday

AI Architecture Thinking

Review:

- architecture,
 - tradeoffs,
 - reliability,
 - scaling.
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WHAT TO AVOID

Avoid becoming:

- prompt-only engineer,
- AI wrapper developer,
- copy-paste AI builder,
- tutorial-dependent engineer.

Focus on:

- systems,
 - architecture,
 - reliability,
 - infrastructure,
 - production thinking.
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KEY TO LONG-TERM SUCCESS

The strongest AI engineers will NOT necessarily be:

- the best ML researchers,
- or the best prompt writers.

They will be engineers who can:

- integrate AI into systems,
- operate AI reliably,
- debug AI workflows,
- scale AI platforms,
- and architect intelligent production systems.

That is the exact purpose of this book.

FINAL MESSAGE

This curriculum is intentionally deep.

Do not rush through it.

The goal is not:

- quick completion,
- certificate collection,
- shallow AI hype participation.

The goal is:

Becoming a deeply capable AI-native systems engineer.

If followed seriously, this book alone can significantly elevate your career relevance in the AI-driven software industry over the next decade.